MLPC Report

Team Quantized Encoders

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1 Baseline Reproduction

1.1 Baseline Training and Prediction Results

The baseline approach achieves Macro F1 score of 0.317 on the non-hidden test set. The class-wise results differ strongly, indicating differences in difficulty of detection. The best performing classes are "running_water", "phone_ringing" and "vacuum_cleaner" with F1 scores of 0.578, 0.485 and 0.470 respectively. The weakest classes are "window_open_close", "wardrobe_drawer_open_close" and "door_open_close" with F1 scores of 0.061, 0.120 and 0.177 respectively. A complete overwiev can be seen in Figure 1.

1.2 Performance Differences between Classes

Event duration is a reason for these differences. Classes with higher F1 scores are typically longer sound events. This can also be seen in Figure 2. This makes them easier to detect with a segment based classifier, since they cover multiple consecutive segments.

The weakest classes in contrast have rather short durations, making them difficult to learn and localize correctly. Also the weakest classes have similar sound patterns and are therefore not easy to distinguish.

The class frequency distribution also contributes to the observed differences. More frequent classes provide more positive examples. This improves learning for the corresponding tree classifier. The class distribution can be seen in Figure 3. This explains especially "running_water", however, "footsteps" and "keyboard_typing" are also very frequent classes and perform worse, indicating that the final performance is a mix of frequency, duration and sound uniqueness.

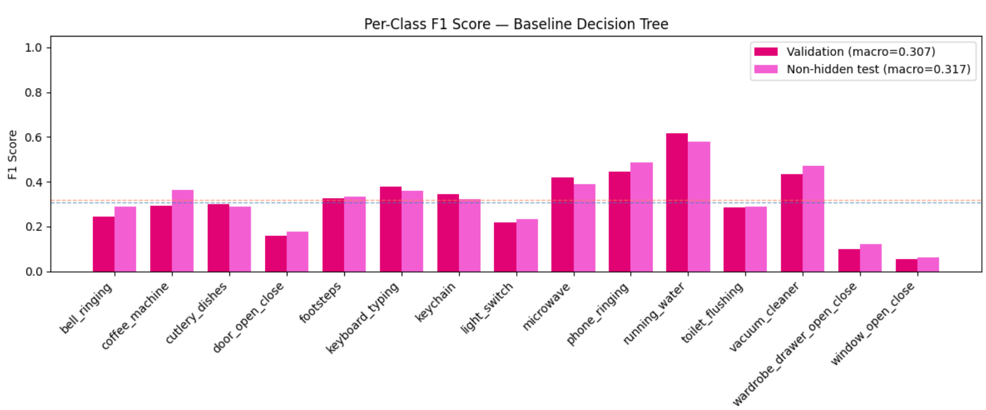

Figure 1: Macro F1 scores per class produced by the given baseline

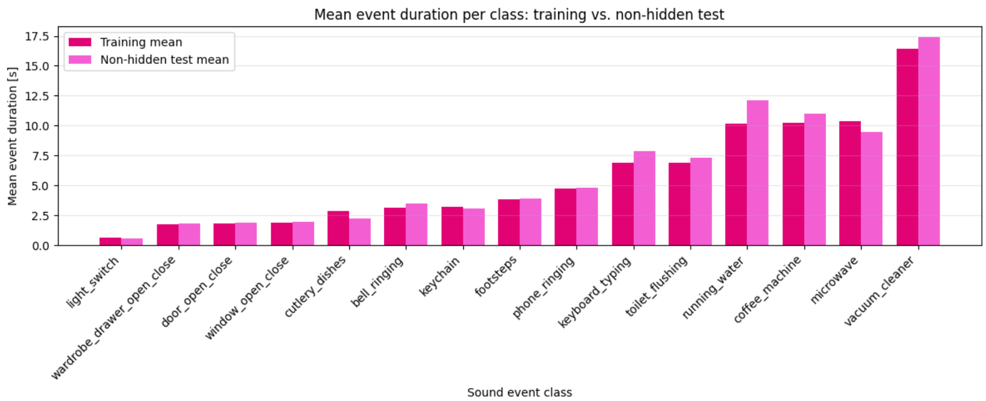

Figure 2: Mean event duration time per class for the training and the non-hidden test set

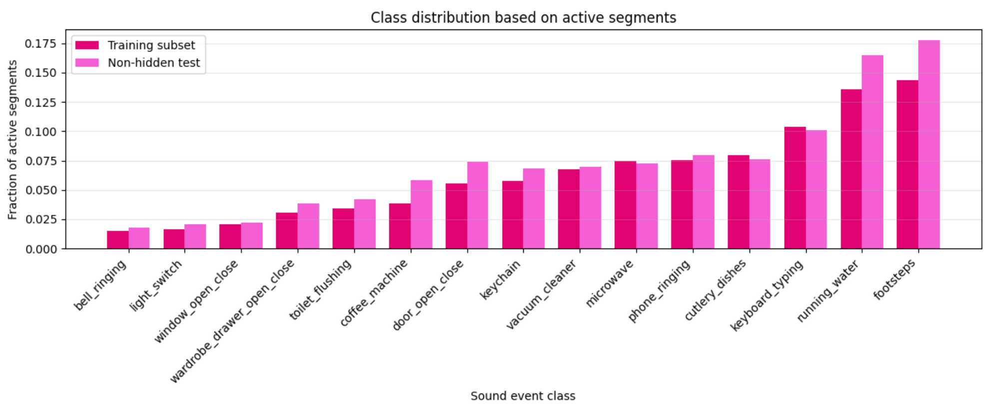

Figure 3: Class frequency distribution based on the 50,000 maximum training segments chosen randomly

1.3 Strengths and Limitations of the Baseline

The baseline has the advantage of being simple and interpretable while remaining computationally efficient. The segment-level classification setup makes this system easy to train and evaluate. Using multiple separate classifiers supports the multi-label setting. Feature representation is also a strength of this approach. The provided features include MFCCs, mel-spectogram statistics and other aggregated audio features. Said features describe important acoustic properties, which are sufficient for some acoustically distinctive and long-lasting classes.

The same feature representation is also limited because it aggregates short features into 1-second statistics, removing much of the detailed temporal structure inside a segment. As a result, discriminative information can be diluted by background noise or overlapping sounds, which could contribute to the low F1 scores of the shorter classes.

Using independent binary classifiers is useful for multi-label prediction, however it ignores dependencies between classes. In the full recordings, events may co-occur or be acoustically or semantically related. Since the classifiers are trained independently, this baseline approach cannot use these relationships.

The decision trees themselves are a limitation, as single trees are often unstable and provide limited ability to capture complex decision boundaries in high-dimensional feature spaces. This might lead to overfitting on some feature thresholds and failing to generalize.

Overall the baseline works well for long, frequent and acoustically distinctive sound events. The biggest limitations are loss of fine temporal information, independent segment-wise predictions, independent class models and limited expressive power of single trees. These limitations also explain the poor performance on short events and on acoustically ambiguous classes.

2 Simple Classifiers

2.1 Logistic Regression as Baseline Replacement

We replaced the baseline's binary decision-tree classifiers with one-vs-rest Logistic Regression (LR) classifiers. LR was selected as the strongest suitable classical model from the previous phase; the CNN was excluded because it is a deep-learning model. LR is efficient, interpretable, compatible with standardized acoustic features, and produces per-class probabilities for multi-label decisions.

We investigated the regularization strength C and class weighting using segment-based Macro F1 on the validation set. The complete sweep is shown in Table 1. The best configuration used $C = 0.01$ and `class_weight=None`. Balanced class weighting reduced Macro F1 for all tested values of C, indicating that any recall gains were outweighed by additional false positives. The final system additionally used validation-tuned per-class thresholds. It achieved a validation Macro F1 of 0.5227 and Micro F1 of 0.6069, and a non-hidden test Macro F1 of 0.5406 and Micro F1 of 0.6199.

Table 1: Validation Macro F1 for different Logistic Regression regularization strengths and class-weighting strategies.

C	`class_weight=None`	`class_weight=balanced`
0.01	**0.5227**	0.4974
0.10	0.5191	0.4929
1.00	0.5195	0.4880
10.00	0.5175	0.4875
30.00	0.5170	0.4874

2.2 Comparison with the Decision-Tree Baseline

On the non-hidden test set, the decision-tree baseline achieved a Macro F1 of 0.3170. The selected LR system increased this to 0.5406, corresponding to an absolute improvement of 0.2236 (Table 2). This shows that regularized linear decision boundaries on standardized features, together with per-class threshold tuning, substantially improve segment-level sound-event detection.

The largest gains were obtained for `keyboard_typing`, `microwave`, `toilet_flushing`, `cutlery_dishes`, and `keychain`, as shown in Table 2. These classes contain relatively stable or repeated acoustic patterns across consecutive segments.

In contrast, `window_open_close`, `light_switch`, and `wardrobe_drawer_open_close` remain challenging (Table 2). These short events may occupy only part of a one-second segment, have imprecise boundaries, or overlap acoustically with other household sounds.

Table 2: Selected class-wise non-hidden test F1 comparison.

Class	Decision Tree	LR	Difference
`keyboard_typing`	0.359	0.713	+0.354
`microwave`	0.390	0.710	+0.321
`toilet_flushing`	0.290	0.575	+0.285
`cutlery_dishes`	0.287	0.558	+0.271
`keychain`	0.321	0.578	+0.256
`door_open_close`	0.177	0.419	+0.242
`wardrobe_drawer_open_close`	0.120	0.357	+0.237
`window_open_close`	0.061	0.217	+0.156
`light_switch`	0.234	0.222	-0.012
Overall Macro F1	0.317	0.541	+0.224

Figure 4 confirms that LR improves most sound event classes, while very short transient events remain difficult in absolute terms.

2.3 Qualitative Error Analysis

To analyse the temporal behaviour of the final Logistic Regression system, we inspect three representative non-hidden test recordings that illustrate a strong success case, a severe false-positive case, and a mixed confusion case.

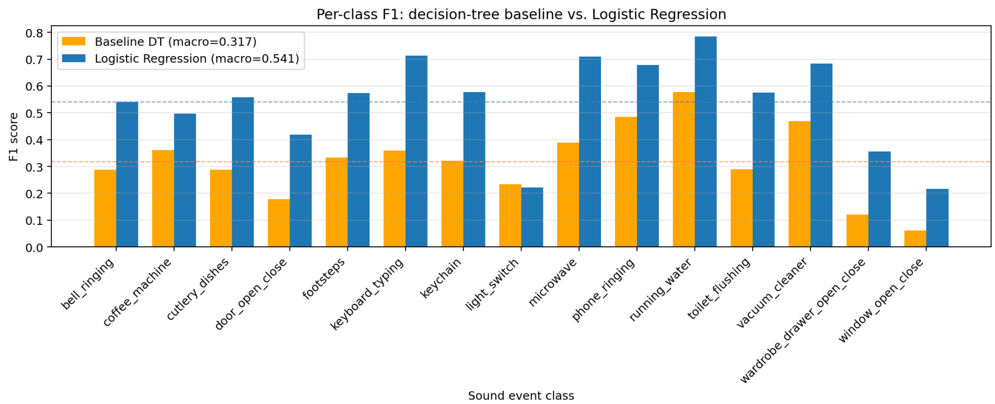

Figure 4: Per-class F1 comparison between the decision-tree baseline and the selected Logistic Regression system. Dashed horizontal lines indicate the respective overall Macro F1 scores.

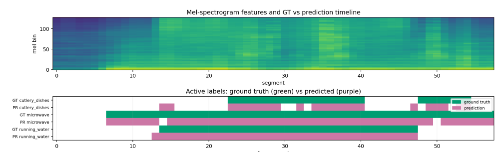

Figure 5: Qualitative success case for `003712.wav`.

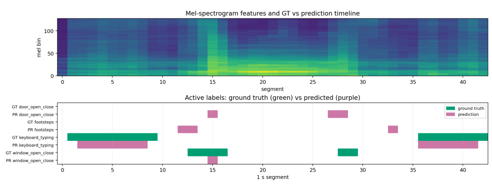

Figure 6: Qualitative confusion case for `003866.wav`.

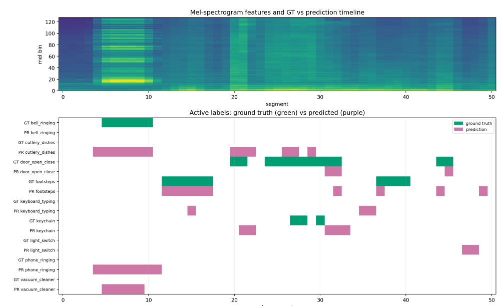

Figure 7: Qualitative failure case for `005541.wav`.

For each recording, we visualize the spectrogram together with the ground-truth event intervals (green) and the predicted event intervals (purple).

Figure 5 shows a strong success case for `003712.wav`, which achieved a file-level micro F1 of 0.949. Both ground truth and predictions contain `cutlery_dishes`, `microwave`, and `running_water`. The prediction tracks the annotated events closely, with mostly overlapping ground-truth and predicted intervals and only small boundary deviations. This example shows that the LR system works well for longer and acoustically stable sound events.

Figure 6 presents a mixed confusion case for `003866.wav`, which achieved a file-level micro F1 of 0.667. The ground truth contains `keyboard_typing` and `window_open_close`, while the predictions additionally include `door_open_close` and `footsteps`. This example highlights confusions between acoustically similar household events, especially `window_open_close` versus `door_open_close`. It also shows that even when the correct classes are detected, extra spurious activations may still reduce the final score.

Figure 7 shows a severe failure case for `005541.wav`, with a file-level micro F1 of only 0.247. The ground truth contains `bell_ringing`, `door_open_close`, `footsteps`, and `keychain`, but the model additionally predicts several incorrect classes, including `cutlery_dishes`, `keyboard_typing`, `light_switch`, `phone_ringing`, and `vacuum_cleaner`. The visualization is dominated by many orange activations without corresponding blue ground-truth intervals. This indicates that LR reacts to local acoustic bursts or short feature patterns, but has no explicit temporal event model to suppress implausible event sequences.

Overall, the qualitative analysis confirms the quantitative findings: the LR system performs well on longer, stable acoustic patterns, but remains sensitive to short transient sounds, acoustically similar events, and temporally noisy segment-level predictions.

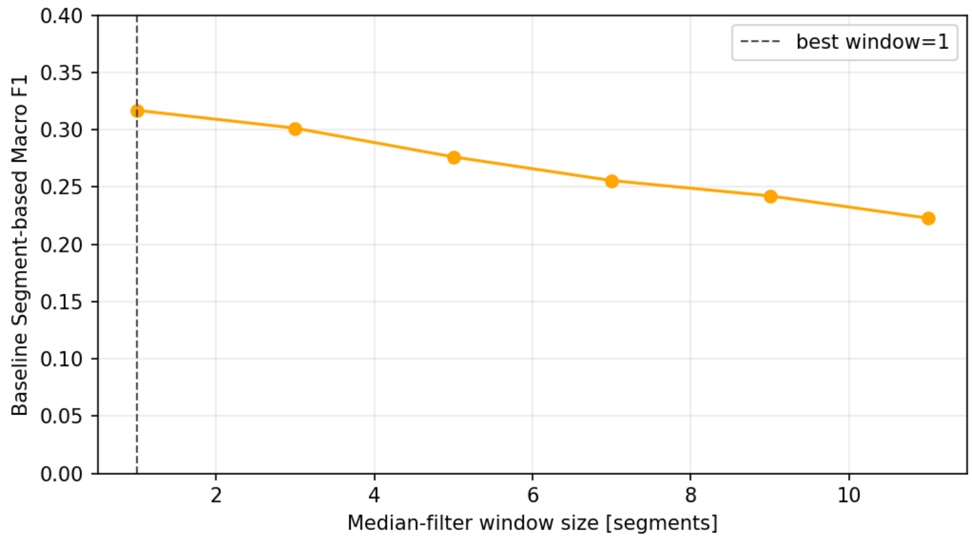

Figure 8: Parameter sweep on window size for baseline performance

3 Post-Processing

3.1 Implementation of Median Filtering

We chose to implement Median Filtering for this task. The investigated parameter is the window size. For this a list was created from 1 to 12 using a step size of 2, meaning only odd values are considered. Note that a window of size 1 means that this is the baseline, where no post-processing happens.

3.2 Performance Comparison

The baseline does not benefit from median filtering. Figure 8 shows that the peak is already reached using a window size of 1, meaning the unprocessed results already give the highest Macro F1 score. For this reason, a class-wise comparison was not computed. The results of the unprocessed baseline approach can be seen in Figure 1.

Median filtering improved the non-hidden-test set of the LR system. The parameter sweep shows that small temporal refinement is helpful, but too much is harmful. The peak was reached using a window size of 5, where the Macro F1 score improved to 0.5591. Larger window sizes reduced performance again. This is shown by Figure 9.

In Figure 10 we have the class-wise comparison of performance with and without post-processing. We can observe that most classes improve performance with median filtering. One possible reason for this could be that the original classifier produced short gaps or short false activations that are then corrected by looking at the neighbors.

4 Reflection and Real-World Considerations

4.1 Technical Limitations

The final LR system still has important limitations. First, predictions are made from one-second aggregated features, which can blur short events and reduce temporal precision. This is visible for `light_switch`, `window_open_close`, and `wardrobe_drawer_open_close` (Table 2). Shorter overlapping windows or frame-level temporal models could improve onset and offset detection.

Second, the classifiers predict each class and segment independently. They do not model event duration or temporal consistency. Figure 7 shows many isolated false-positive activations, while Figure 6 shows confusion between `window_open_close` and `door_open_close`. Possible improvements include class-specific smoothing, duration constraints, or sequence models operating on consecutive segments.

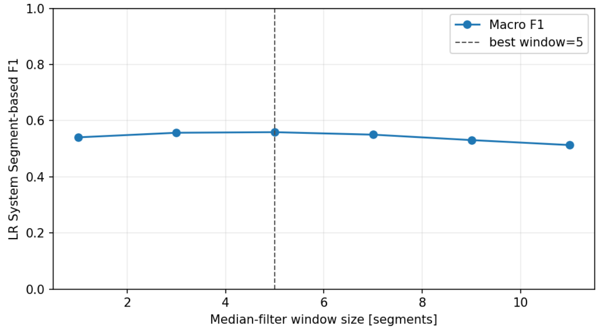

Figure 9: Parameter sweep on window size for LR performance

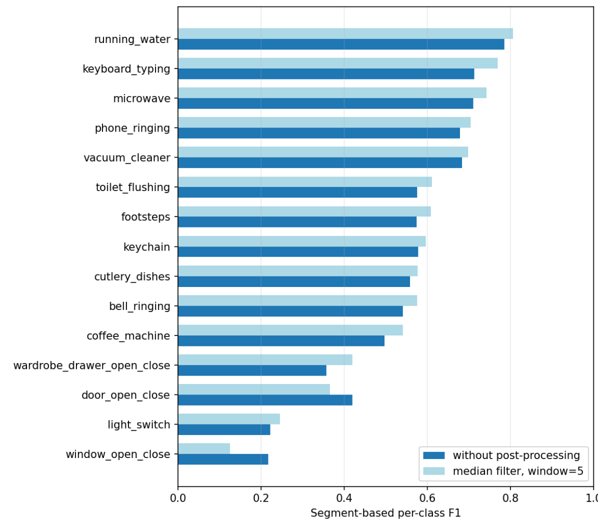

Figure 10: Class-wise comparison of LR with and without post-processing

Finally, Logistic Regression can only learn linear decision boundaries. Non-linear models such as Random Forests, XGBoost, or compact neural models could better model interactions between acoustic features. More varied training data and augmentation with room noise, reverberation, and different microphone positions would also improve robustness.

4.2 Real-World Application

The current system is suitable as a lightweight prototype, but not yet for reliable autonomous smart-home control. Although LR reaches a Macro F1 of 0.5406, the qualitative examples still show false detections, confusions, and timing errors. In real homes, performance would likely decline because of different rooms, microphones, appliances, background noise, and overlapping events.

Inference is computationally inexpensive and could run locally on CPU hardware. However, the one-second segmentation introduces latency, and the median filter with a five-segment window adds further delay when future context is used. This is acceptable for activity logging, but not for immediate alerts.

A realistic first deployment would therefore be low-risk activity monitoring or event summaries. Before using the system for notifications or automation, it would require broader real-home evaluation, user-specific threshold calibration, an uncertainty or “unknown event” option, and privacy-preserving local audio processing.

5 Disclosure of LM and AI Tool Use

Throughout the creation of this report, Large Language Models (LLMs) were used to support various stages of the workflow, especially coding and improving the clarity and quality of our writing.

For coding parts we used Claude Code as well as ChatGPT Thinking. For writing improvements mainly ChatGPT Thinking was used.

All generated code suggestions and textual suggestions were checked manually. Code was verified by running the corresponding notebooks, inspecting the produced outputs, and comparing the reported numbers with the saved experiment results. Textual suggestions were only included if they matched our own experiments and interpretations.